To	**The Maintainers W.L.L., Marina Tower 50, 11th Floor, P.O. Box 201325, Lusail, Qatar**
Attn	Mr. Fahad Poyil, Procurement Specialist
From	Metri Engineering Services W.L.L. (MES), Doha, Qatar
Our Ref	QREF 04/11/2026-1
Your Ref	RFQ dated 26 Oct 2026, Contract No. QIA-MRC-2025-001, Section 5.1.1, Package 1, Item C
Project	Qatar Investment Authority Office, Ooredoo Tower (Building 14), Levels 02 to 14
Subject	Technical and Commercial Proposal: Airflow Balancing of Floor AHU Distribution System and Verification of VAV Boxes

Dear Mr. Fahad,

MES is pleased to submit its technical and commercial proposal for airflow balancing and VAV verification and recalibration at QIA Levels 02 to 14, at a lump sum of **QAR 140,000 per visit** (QAR 280,000 per year for two visits).

1. Introduction

MES provides testing, adjusting and balancing (TAB), performance assessment and maintenance services for commercial towers, hotels, hospitals and industrial facilities in Qatar.

2. Project Basis and Asset Base

The works cover 13 floors (Levels 02 to 14), each served by its floor AHU, with 391 VAV boxes per the latest drawings. Two visits are carried out per year: May (start of summer) and October (start of winter).

Floor	VAV Boxes (Nos)	Floor AHU (Nos)	Fresh Air VAV + CO_2 Sensor (Nos)
Level 02	23	1	1
Level 03	38	1	1
Level 04	34	1	1
Level 05	36	1	1
Level 06	30	1	1
Level 07	34	1	1
Level 08	37	1	1
Level 09	1	1	1

Floor	VAV Boxes (Nos)	Floor AHU (Nos)	Fresh Air VAV + CO_2 Sensor (Nos)
Level 10	32	1	1
Level 11	38	1	1
Level 12	39	1	1
Level 13	25	1	1
Level 14	24	1	1
Total	**391**	**13**	**13**

Air outlets (supply and return linear slot diffusers, grilles, diffusers and registers) within the scope areas of Levels 02 to 14 are covered as **one lot (lump sum)**.

3. Detailed Scope of Work

MES will carry out testing, adjusting and balancing of the floor air distribution systems (AHUs, VAV boxes and air outlets) on Levels 02 to 14, per visit, as follows.

1. **Pre-visit review and site survey**
   - Review of as-built drawings, VAV schedules, MVAC Operations Manuals and previous TAB/traverse records.
   - Joint walk-down with the client to confirm scope areas, access, ceiling types and BMS points.
2. **AHU airflow measurement (13 Nos)**
   - Measurement of supply, return and fresh air at each floor AHU, including duct traverse (pitot) measurements where applicable.
   - Recording of fan and motor data: external and total static pressures, motor running current and voltage, VFD frequency and fan speed.
   - Comparison of measured flows against the design and the consultant-approved airflow basis.
3. **airflow balancing**
   - balancing of the MVAC distribution system on each floor, including the fresh air supplied to each floor AHU.
   - Adjustment of branch and volume dampers to achieve the design flow distribution within the available AHU airflow.
4. **VAV box verification and recalibration (391 Nos)**
   - Measurement of actual airflow at each VAV box at maximum and minimum set-points, driven through the BMS.
   - Comparison of measured flow against the BMS-indicated flow
   - Functional verification of VAV damper/actuator operation and thermostat response.
5. **Fresh air VAV boxes (13 Nos)**
   - Verification of the fresh air VAV box serving each floor AHU
6. **CO_2 sensor testing (13 Nos)**
   - Functional test of each fresh-air CO_2 sensor against a calibrated reference CO_2 meter.

- Verification of the sensor reading at the BMS and of the fresh air VAV response to CO_2 level (BMS side).

7. **Air terminal measurements (1 lot)**
   - Measurement of airflow at each supply and return grille, diffuser and air terminal within the scope areas.
   - Rebalancing of the ductwork distribution where high or low airflow is identified.
8. **Performance verification**
   - Identification and reporting of operational issues involving AHUs, VFDs, low airflow, dampers, actuators and ductwork leakage (visual and audible identification).
   - Reporting of issues requiring landlord intervention and coordination with the client accordingly.
9. **Testing and commissioning reports**
   - Floor-wise TAB reports in NEBB format.

4. Methodology and Reporting

Balancing basis. Airflow will be balanced proportionally to the available airflow of each floor AHU, on the design airflow basis approved by the consultant/as-built for each floor. Where AHU airflow or fresh air is below design, the shortfall will be reported with recommendations.

Floor sequence per visit:

1. AHU baseline measurements (supply, return, fresh air, static pressures, VFD).
2. Fresh air VAV and CO_2 sensor verification.
3. VAV zones: VAV at maximum/minimum, supply outlet readings, damper adjustment, recalibration.
4. Return outlet measurements.
5. Final AHU readings after balancing and defects log.

Reporting. A NEBB-format TAB report will be submitted for each floor, consolidated into a final report per visit. Each report includes:

- Measured readings and test sheets for AHUs, VAV boxes, fresh air VAVs, CO_2 sensors and air outlets.
- Design versus actual comparison with percentage deviation.
- Balancing results and VAV recalibration details (before/after flows, correction factors, set-points).
- List of instruments used with calibration certificates.
- Identified defects (AHU, VFD, dampers, actuators, ductwork leakage, low airflow) with corrective recommendations.

5. Client Side

The following are to be provided by the client (The Maintainers / QIA) at no cost to MES during each visit:

1. **BMS operator and technician** present during all MES working shifts, to drive VAV boxes to maximum/minimum, enter recalibration factors and set-points, and confirm CO_2 sensor readings and BMS interface.
2. **Coordination with the landlord (Ooredoo)** for operation of AHUs, VFDs and chilled water supply during out-of-hours testing, with systems running at normal operating conditions.
3. **Filters replaced or cleaned and coils cleaned** before each visit, so that readings represent normal operating conditions.
4. **Site access, security clearance, gate passes and escorts** for MES personnel and instruments for out-of-hours work.
5. **Latest as-built drawings, VAV schedules, MVAC Operations Manuals and approved design airflow basis** per floor.
6. **Clear access to ceilings** in occupied areas, including removal of furniture or obstructions where required.
7. **Work permits** and nomination of a client representative to witness and approve readings.

6. Exclusions

The following are excluded from this proposal:

1. **Ductwork repairs.** Leakage and defects identified will be reported; any required ductwork repairs will be quoted separately after submission of the report.
2. **Duct leakage pressure testing** (DW/143 or equivalent).
3. **BMS operator, technician and BMS programming**, including software, graphics, controller replacement or reconfiguration beyond entry of recalibration factors and set-points.
4. **Supply, replacement or repair** of VAV boxes, actuators, controllers, thermostats, CO_2 sensors, dampers, diffusers, grilles or any HVAC equipment or components.
5. **Mechanical and electrical repairs** to AHUs, fans, motors, VFDs, belts, pulleys and bearings, including lubrication and electrical works.
6. **Rectification of AHU capacity or fresh air shortfalls.** Where AHU or fresh air flow is below design, balancing is proportional to the available airflow and the shortfall is reported.
7. **Removal and cleaning of filters, coils and strainers.**
8. **Scaffolding and powered access equipment** (aerial work platforms, scissor lifts). Ladders and podium steps for ceiling-level access are included.
9. **Exhaust systems** (toilet, pantry and kitchen exhaust), fan coil units, power induction units (PIUs), chilled water and other hydronic systems.
10. **Areas outside the scope areas** shown on the as-built drawings for Levels 02 to 14.
11. **Gypsum or fixed ceilings without access panels**; cutting, making good or providing access panels.
12. **Design verification or redesign** of the air distribution system.
13. **Indoor air quality, noise or vibration testing** beyond the scope stated.
14. **Re-testing or additional visits** caused by system non-readiness, landlord plant unavailability, lack of BMS support, access restrictions or design changes; these will be charged separately.
15. **Any works not explicitly described** in Section 3.

7. Commercial Offer

The lump sum price is **QAR 140,000 per visit**, or **QAR 280,000 per year** for two visits (May and October).

No.	Description	Unit	Qty	Amount (QAR)
1	Air handling units: airflow measurement (SA/RA/FA), duct traverses and performance check	Nos	13	Included
2	VAV boxes: airflow verification and recalibration	Nos	391	Included
3	Fresh air VAV boxes: verification and recalibration	Nos	13	Included
4	CO_2 sensors: testing and BMS interface verification	Nos	13	Included
5	Air outlets (grilles, diffusers, linear slot diffusers, registers): measurement and balancing	Lot	1	Included
6	Proportional airflow balancing of floor distribution systems, Levels 02 to 14	Lot	1	Included
7	Ceiling tile removal and reinstatement, access ladders, floor protection	Lot	1	Included
8	MOS, RA, floor-wise and final NEBB-format TAB reports	Lot	1	Included
	Lump sum price per visit			**140,000**
	Total annual price (2 visits)			**280,000**

All prices are in Qatari Riyals.

9. Payment Terms and Conditions

Payment terms (per visit)

Milestone	Percentage
Upon purchase order / start of each visit	40%
Upon completion of site TAB field activities	40%
Upon submission of the final TAB report	20%

Terms and conditions

1. **Validity:** 30 days from the date of this proposal.
2. **Duration:** approximately 30 working days on site per visit, out of office hours, plus 10 working days for the final report, subject to system readiness and site access.
3. **Visit schedule:** two visits per year (May and October). The first visit date will be agreed upon award.

4. **System readiness:** works depend on AHUs, VFDs, chilled water and BMS being operational during the scheduled working hours.
5. **Additional visits:** any visit beyond the two scheduled visits will be priced at the per-visit rate, pro-rata to the floors covered.
1. **Defects and repairs:** identified defects, including ductwork repairs, will be quoted separately after the report.
2. **Taxes and duties:** prices exclude any taxes or duties that may be introduced by the authorities after the date of this proposal.

METRI ENGINEERING SERVICES

Fayez METRI (55572793 / 70500311)

